25 ALKENES

The alkenes are another family of hydrocarbons. They all contain a carbon–carbon double bond. Alkenes can be produced from alkanes by cracking and they are much more reactive than the alkanes.



▶ Figure 25.1 Sunflower oil is polyunsaturated and contains lots of C=C groups.

LEARNING OBJECTIVES

- Know that alkenes contain the functional group >C=C<
- Know the general formula for alkenes
- Explain why alkenes are classified as unsaturated hydrocarbons
- Describe the reactions of alkenes with bromine to produce dibromoalkanes
- Describe how bromine water can be used to distinguish between an alkane and an alkene

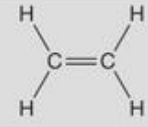
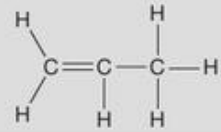
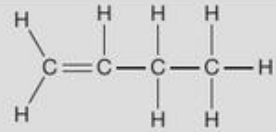
The alkenes are a homologous series of hydrocarbons which contain a carbon–carbon double bond. The **C=C** bond is the functional group of the alkenes.

The first three members of the alkene series are shown in Table 25.1.

KEY POINT

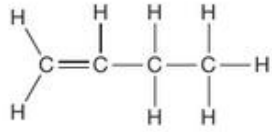
It is not possible to have an alkene with one carbon atom because alkenes must contain a C=C group so you need a minimum of two carbon atoms.

Table 25.1 The first three alkenes

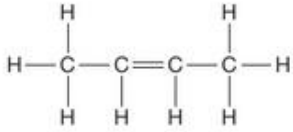
Name	Molecular formula	Empirical formula	Structural formula	Displayed formula
ethene	C_2H_4	CH_2	$CH_2=CH_2$	
propene	C_3H_6	CH_2	$CH_2=CHCH_3$	
but-1-ene	C_4H_8	CH_2	$CH_2=CHCH_2CH_3$	

ISOMERS

There are two straight-chain isomers with the formula C_4H_8:

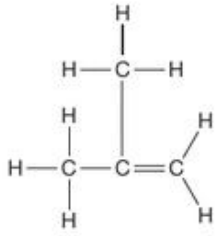


but-1-ene



but-2-ene

and one branched-chain isomer:



HINT

A way of thinking about this is that the alkenes are not saturated with H atoms, they do not have the maximum number of H atoms for that number of C atoms.

UNSATURATED HYDROCARBONS

Alkenes are *unsaturated* compounds because they contain a C=C bond. *A saturated compound contains single C–C bonds only whereas an unsaturated compound contains one or more double or triple C–C bonds.*

THE GENERAL FORMULA

Alkenes have the general formula $\mathbf{C_nH_{2n}}$. The number of hydrogen atoms is twice the number of carbon atoms. This means that all alkenes have the same empirical formula, CH_2.

The alkene with 11 carbon atoms will have 22 hydrogen atoms and the molecular formula $C_{11}H_{22}$.

PHYSICAL PROPERTIES

These are very similar to those of the alkanes. Remember that the small alkanes with up to four carbon atoms are gases at room temperature. The same is true for the alkenes. They are gases up to C_4H_8, and the next dozen or so are liquids. Again, the members of the homologous series *show a trend in physical properties*.

CHEMICAL REACTIONS OF THE ALKENES

COMBUSTION

In common with all hydrocarbons, alkenes burn in air or oxygen to give carbon dioxide and water, for example:

$$C_2H_4(g) + 3O_2(g) \rightarrow 2CO_2(g) + 2H_2O(l)$$

This isn't a reaction anybody would ever choose to do. Alkenes are much too useful to waste by burning them.

ADDITION

The C=C double bond is the functional group. The functional group determines the chemical properties of a compound. Alkenes have characteristic chemical properties which are different from those of alkanes. Alkenes are much more reactive than alkanes.

Alkenes undergo addition reactions. Part of the double bond breaks to become a single C–C bond and the electrons are used to join other atoms onto the two carbon atoms.

THE ADDITION OF BROMINE

Bromine adds to alkenes without any need for heat, light or a catalyst. The reaction is often carried out using bromine water (aqueous bromine solution) as, for example, in Figure 25.2, with ethene.

HINT

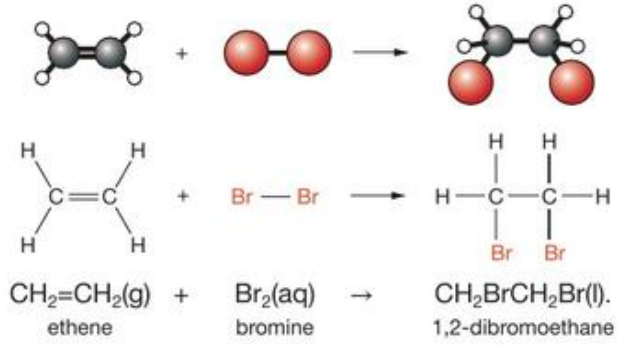
Decode the name and make sure you can see how it works. 'eth' shows a two-carbon chain. 'ane' shows only carbon–carbon single bonds. The beginning of the name, '1,2-dibromo', shows two bromine atoms attached to the first and second carbon atoms.

You know a reaction has happened because bromine water is *orange* but the product, called 1,2-dibromoethane, is a *colourless* liquid.

You can write this as an equation in two ways: first, using displayed formulae and second, using structural formulae.

HINT

Make sure you can see how these formulae relate to the full structures in the equation above.



▲ Figure 25.2 An addition reaction: an alkene plus bromine

THE TEST FOR UNSATURATED COMPOUNDS

Any compound with a carbon–carbon double bond will react with bromine in a similar way to ethene. This is used to *test for a carbon–carbon double bond*. If you shake an unknown organic compound with bromine water and the orange bromine water is *decolourised* (the colour changes from orange to colourless), the compound contains a carbon–carbon double bond. If your unknown compound is a gas, you can simply bubble it through bromine water with the same effect.

HINT

You must be careful with the words you use here: the final mixture is *colourless* and not clear. Do not use the word clear. Clear means not cloudy, you can see through it. The bromine water is clear and any other solution can also be described as clear.

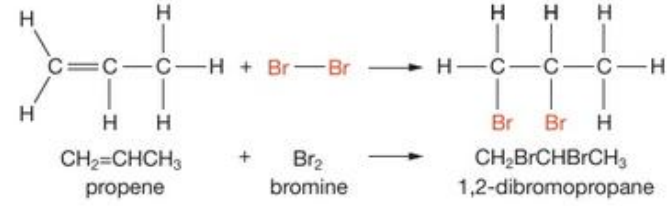
The left-hand tube in Figure 25.3 shows the effect of shaking a liquid alkene with bromine water. The organic layer (containing the alkene) is on top. You can see that the bromine water has been completely decolourised, showing the presence of the carbon–carbon double bond.



▲ Figure 25.3 The result of shaking a liquid alkene (left) or alkane (right) with bromine water.

The right-hand tube in Figure 25.3 shows what happens if you use a liquid alkane, which doesn't have a carbon–carbon double bond. The colour of the bromine is still there. Alkanes do not decolourise bromine water because they do not contain a carbon–carbon double bond.

The addition reaction between bromine and propene is very similar to that of ethene. Part of the carbon–carbon double bond breaks and two bromine atoms add across the bond, as shown in Figure 25.4.



▲ Figure 25.4 An addition reaction of propene with bromine

The product is called 1,2-dibromopropane. Its structure is based on propane and the prefix '1,2-dibromo' shows there are two bromine atoms, on the first and the second carbon atoms.

EXTENSION WORK

In Figure 25.3, interestingly, most of the colour in the right-hand tube is now in the top organic layer. That is because the covalent bromine, which has a simple molecular structure, is more soluble in the organic compound than it is in water.

OTHER IMPORTANT ADDITION REACTIONS OF ALKENES



▲ Figure 25.5 Production of margarine

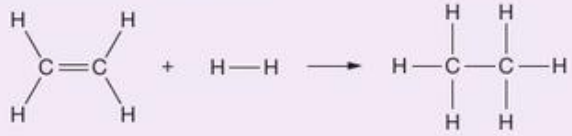
EXTENSION WORK

The addition of water to propene could produce two different products, propan-1-ol and propan-2-ol. In reality, much more propan-2-ol is formed than propan-1-ol. In the major reaction, a hydrogen atom from water is first added onto the end carbon and then the –OH group is attached to the middle carbon. The addition of compounds to an unsymmetrical alkene like propene will be explored in detail when you study chemistry at a more advanced level.

EXTENSION WORK

Hydrogenation

A **hydrogenation** reaction involves the *addition of hydrogen*. An alkene reacts with hydrogen to form an alkane, for example:



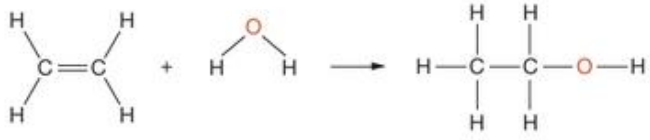
$CH_2{=}CH_2 + H_2 \rightarrow CH_3CH_3$

This reaction is used for the production of margarine, which is made by hydrogenation of polyunsaturated (i.e. many C=C bonds) plant oils.

CHEMISTRY ONLY

HYDRATION

A **hydration** reaction involves the *addition of water*. An alkene reacts with water to form an alcohol, for example:



$CH_2{=}CH_2 + H_2O \rightarrow CH_3CH_2OH$

Starting materials:	ethene and steam
Temperature:	300 °C
Pressure:	60–70 atmospheres (the atmosphere is a unit of pressure, equal to 101325 pascals)
Catalyst:	phosphoric acid

This reaction is used for the production of high-purity ethanol in industry (see Chapter 26 for more detail on this reaction).

END OF CHEMISTRY ONLY

CHAPTER QUESTIONS

SKILLS CRITICAL THINKING

1 The alkenes are a *homologous series* of *unsaturated* hydrocarbons.

a i State two characteristics of the members of a *homologous series*.

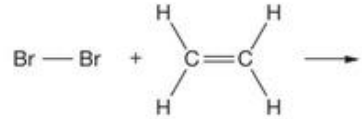
ii Explain the term *unsaturated*.

b Ethene is the simplest member of the alkenes. Bromine water can be used to distinguish alkenes from alkanes.

i Bromine water is added to ethene. State the starting colour of the bromine water and the finishing colour of the reaction mixture.

SKILLS PROBLEM SOLVING, INTERPRETATION

ii Complete the equation by drawing and naming the displayed formula of the product.



Name of the product:

SKILLS INTERPRETATION 8

c Isomers are compounds that have the same molecular formula but different displayed formulae. Draw the displayed formulae of three alkene isomers that have the molecular formula C_4H_8.

d Can you draw any other isomers of C_4H_8 which are *not* alkenes?

SKILLS CRITICAL THINKING 6

2 A gaseous hydrocarbon, **P**, with three carbon atoms, decolourises bromine water.

a Explain what the decolorisation of the bromine water tests for.

7

b Write the displayed formula for the hydrocarbon.

SKILLS PROBLEM SOLVING 10

c Write a balanced chemical equation for the reaction between the hydrocarbon and bromine, using displayed formulae.

d The reaction between ethane and bromine gas in the presence of UV light also causes the bromine to lose colour.

9

i Write a balanced chemical equation for this reaction.

SKILLS REASONING 7

ii This reaction is described as a substitution reaction, whereas the reaction between **P** and bromine is an addition reaction. Using the equations you have already written, explain the difference between addition and substitution.

SKILLS CRITICAL THINKING 8

3 Alkenes undergo addition reactions.

a Predict the structural formula of the compound formed when chlorine reacts with propene. Name the product formed.

b Ethene reacts with hydrogen. Give the name and structural formula of the product.

6

4 Propene can be made by heating propane and sulfur.

a Name another method of making propene and state two conditions for making alkenes from alkanes.

SKILLS EXECUTIVE FUNCTION

b Describe a chemical test which can be used to distinguish between propane and propene.

Test:

Result with propane:

Result with propene:

SKILLS CRITICAL THINKING 8

5 Give the structural formulae and names of the product when but-1-ene reacts with each of the following:

a steam

b hydrogen chloride (HCl)

c bromine.